ISyE 6669
2019 Fall Midterm Solution

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Problem 1. Consider the matrix $A = \begin{bmatrix} 0 & 3 & -2 & -1 \\ 4 & -9 & 1 & 3 \\ -6 & -15 & -1 & 5 \end{bmatrix}$. The rank of A is

A 1

B 2

C 3

D 4

Solution: C Clearly rank(A) ≤ 3. By applying Gaussian elimination on the rows of A we get:

$$\begin{bmatrix} 0 & 3 & -2 & -1 \\ 4 & -9 & 1 & 3 \\ -6 & -15 & -1 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 3 & -2 & -1 \\ 0 & -19 & \frac{1}{3} & \frac{19}{3} \\ -6 & -15 & -1 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & -\frac{37}{19} & 0 \\ 0 & -19 & \frac{1}{3} & \frac{19}{3} \\ -6 & -15 & -1 & 5 \end{bmatrix}$$

from the last matrix we indeed see that the rank of A is equal to 3 since the row vectors are linearly independent.

Problem 2. Suppose matrices $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{m\times n}$ both have full row rank. Then the rank of $A + B$ is

A m

B n

C $\min\{m, n\}$

D Depends on A and B

Solution: D Consider the special case when $m = n = 2$ and the matrices:

$$A = I, B_1 = I, B_2 = -I$$

All of them have full row rank. However, $A + B_1$ has rank $2 = m = n = \min\{m, n\}$ and $A + B_2$ has rank 0.

Problem 3. Consider the function $f(x, y) = x^3 - 2xy + y^2$. The first order Taylor's expansion of this function at $(-2, 3)$ is

A $6x + 2y - 13$

B $5x + 3y - 6$

C $13x - 3y - 6$

D $6x + 10y - 5$

Solution: D The Taylor expansion at $(-2, 3)$ is given by:

$$f(-2,3) + \nabla f(-2,3)^\top \begin{pmatrix} x+2 \\ y-3 \end{pmatrix} = 13 + \begin{pmatrix} 6 \\ 10 \end{pmatrix}^\top \begin{pmatrix} x+2 \\ y-3 \end{pmatrix} = 6x + 10y - 5$$

Problem 4. The set $X = \{x \in [-1, 1]^5 : \sum_{i=1}^5 x_i \geq -2\}$ is

A Unbounded and closed

B Bounded and not closed

C Bounded and closed

D Unbounded and not closed

Solution: C Clearly X is bounded as a subset of the cube $[-1, 1]^5$ which is a bounded set. Moreover, X is a polyhedron. Thus is closed. The correct answer is C.

Problem 5. Which set is convex

A $\{x \in \mathbb{R}^2 | \min\{x_1, x_2\} \geq 1\} + \{x \in \mathbb{R}^2 | x_1 \leq -1, x_2 \leq -2\}$

B $\{x \in \mathbb{R}^2 | x_1 \geq \ln x_2,\ x_2 > 0\}$

C $\{x \in \mathbb{R}^2 | x_1^2 + x_2^2 \geq 4\}$

D $\{x \in \mathbb{R}^2 | \max\{x_1, x_2\} \geq 1\}$

Solution: A A is a convex set. The function $\min\{x_1, x_2\}$ is concave making the first set in the sum convex. The second set is a polyhedron and thus convex. Convexity is preserved under the Minkowski (set) summation. B,C,D describe sets of the form $\{f(x) \geq c\}$ where f is convex, and thus none of them is convex.

Problem 6. Which function is convex on the indicated domain

A $f(x_1, x_2) = 2^{|x_1+1|+|x_2-x_1|}$ on $\mathbb{R}^2$

B $f(x) = -2x^2 + 6x + 10$ on $\mathbb{R}$

C $f(x) = x^5$ on $[-1, 1]$

D $f(x) = \ln(x^2)$ on $(0, +\infty)$

Solution: A (A) is a convex function as the composition of a convex increasing function and a piecewise linear function. (B), (D) are concave functions and (C) is neither concave nor convex.

Problem 7. Which statement is TRUE?

A A function with convex level sets is always convex.

B An optimal solution of maximizing a convex function over a compact set lies on the boundary of the set.

C A non-convex optimization problem always has a unique optimal solution.

D A convex function cannot be a concave function at the same time.

Solution: B (A) is false. For a counterexample, consider any concave increasing function (e.g $\log(x)$). (B) is correct. (C) is false since non-convex problems may have multiple optimal solutions (e.g $\max\{x^2 : -1 \leq x \leq 1\}$). (D) is false since any affine function is both convex and concave.

Problem 8. Which of the following is a convex optimization problem

A $\min\{x^4 + y^4 : x^2 + y^2 \geq 20\}$

B $\max\{\cos(x) : x \in [0, 2\pi]\}$

C $\min\{\ln(x) : x \geq 1\}$

D $\min\{-x - 8y : x + 7y \geq 10, e^x \leq 1\}$

Solution: D The feasible set of A is non-convex. The objective function of (B) is not concave. The objective function of C is not convex. D is correct since the objective function is affine, the first constraint linear and e^x convex.

Problem 9. What is the outcome of the problem $\max\{412x - 511y \ : \ 0 \leq x \leq 1, y \leq 1\}$

A Infeasible

B Unbounded optimum

C The unique optimal solution is $x = 1, y = 1$.

D None of above

Solution: B Set $x = 0$ and let $y \to -\infty$. Then the objective value goes to $+\infty$ meaning that the problem has an unbounded optimum.

Problem 10. I have solved an optimization problem and got an optimal solution x^*. Suppose now a new constraint is added to my problem. If I find that x^* satisfies the new constraint, then x^* is an optimal solution of the modified problem.

A True

B False

C It depends on whether the optimization problem is maximization or minimization.

Solution: A Without loss of generality assume that we deal with a minimization problem. By adding an additional constraint, the optimal value of the new problem OPT_2, will be lower bounded by the optimal value of the original problem OPT_1 (e.g $OPT_2 \geq OPT_1$), since the new feasible set is a subset of the initial one. Since x^* also satisfies the new constraint and the objective stays the same, we get that $OPT_1 \geq OPT_2$. Combining this with the first inequality, we get that x^* is an optimal solution of the modified problem.

Problem 11. If I maximize a univariate convex function over a nonempty closed and bounded interval, then there has to be an optimal solution which is one of the end points.

A True

B False

Solution: A Let $f(a)$ and $f(b)$ be the function values at the end points of the interval. Then since f is convex, the line segment defined by $\lambda f(a) + (1 - \lambda)f(b)$ for $\lambda \in [0, 1]$ lies above the function, so whichever of $f(a)$ and $f(b)$ is larger will not only be the maximum value for this segment but also for the function.

Problem 12. Consider an unconstrained maximization problem $\max f(x)$. Suppose at a point x_0 we know $\nabla f(x_0) = 0$ and $\nabla^2 f(x_0)$ is positive definite. Then

A x_0 is a local maximizer

B x_0 is a local minimizer

C x_0 may be neither a local maximizer nor a local minimizer

D Whether x_0 is a local maximizer or minimizer depends on whether $f(x)$ is a convex function or not.

Solution: B $\nabla f(x_0) = 0$ tells us we have "flattening out" behavior, and $\nabla^2 f(x_0)$ being positive definite indicates that x_0 must be a local minimum.

Problem 13. The gradient vector of $f(x_1, x_2) = \sqrt{x_1 - 1} + (x_2 - 1)^{1/4}$ at point $(x_1, x_2) = (3, 2)$

A Not defined

B $\left(\frac{1}{2\sqrt{2}}, \frac{1}{4}\right)^\top$

C $\left(\frac{1}{\sqrt{2}}, 1\right)^\top$

D $\left(\frac{\sqrt{2}}{2}, \frac{1}{4}\right)^\top$

Solution: B $\frac{\partial f}{\partial x_1} = \frac{1}{2}(x_1 - 1)^{-\frac{1}{2}}$, $\frac{\partial f}{\partial x_2} = \frac{1}{4}(x_2 - 1)^{-\frac{3}{4}}$, and evaluating these partials at 3 and at 2, respectively, yields $\left(\frac{1}{2\sqrt{2}}, \frac{1}{4}\right)^\top$.

Problem 14. The Hessian matrix of $f(x_1, x_2) = 2(x_1 - 3)^2 + 6x_1x_2 + 2(x_2 + 3)^2 + 6x_1 + 2x_2$ at point $(x_1 = 2, x_2 = 3)$ is

A $\begin{bmatrix} 4 & 6 \\ 6 & 4 \end{bmatrix}$

B $\begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}$

C $\begin{bmatrix} 4 & 8 \\ 8 & 2 \end{bmatrix}$

D $\begin{bmatrix} 4 & 4 \\ 4 & 1 \end{bmatrix}$

Solution: A Differentiate the function with respect to x_1 and x_2, then differentiate each of those derivatives with respect to both x_1 and x_2, and evaluate these expressions at (2,3) to get A.

Problem 15. Consider the problem min $\{\boldsymbol{c}^\top \boldsymbol{x} : \boldsymbol{x} \in X\}$ where X is a nonempty compact set in $\mathbb{R}^n$. Suppose $\boldsymbol{c} \neq \boldsymbol{0}$. What can you say about the problem:

A The problem may have an unbounded optimal solution

B An optimal solution can be in the interior of X

C The problem can be infeasible

D The optimal solution exists and is unique

Solution: B The problem cannot be unbounded because X is compact (and thus bounded). The problem cannot be infeasible since X is non-empty. The following example demonstrates both why B is true and why D is false:
min $x_1 + x_2$
s.t. $x_1 + x_2 = 1$
$0 \leq x_1, x_2 \leq 1$

Problem 16. Which of the following is not a feasible point of $X = \{x \in \mathbb{R}^2 \mid x_1 - x_2 \leq -1, 2x_1 - x_2 \leq 0, x_2 \geq -\frac{1}{2}\}$?

A $(0, -\frac{1}{2})$

B $(\frac{1}{2}, \frac{3}{2})$

C $(0, 1)$

D $(-1, 0)$

Solution: A The selected point is the only one of the four listed that does not fit all constraints.

Problem 17. For the transportation problem:

$$\begin{aligned} \min \quad & \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i=1}^{m} x_{ij} \geq d_j, && j = 1, 2, ..., n \\ & \sum_{j=1}^{n} x_{ij} \leq s_i, && i = 1, 2, ..., m \\ & x_{ij} \geq 0 && i = 1, 2, ..., m, j = 1, 2, ..., n \end{aligned}$$

Which of the following situations must cause infeasibility?

A The total supply $\sum_{i=1}^{m} s_i$ is greater than the total demand $\sum_{j=1}^{n} d_j$

B There exists a demand d_j which is greater than the total supply $\sum_{i=1}^{m} s_i$

C The total supply $\sum_{i=1}^{m} s_i$ is equal to the total demand $\sum_{j=1}^{n} d_j$

D Some s_i is smaller than all demand d_j for $j = 1, 2, ..., n$.

Solution: B Having more supply than demand isn't usually a problem since we're only really concerned with satisfying demand, and having equal supply and demand is fine as long as all supply can travel through the proper channels to meet demand. Having one supply value smaller than all demand values doesn't tell you anything about the rest of the supply values, so this may not cause infeasibility. Having one demand greater than the total supply guarantees infeasibility since that demand cannot possibly be met.

Problem 18. Which of the following functions can be reformulated as convex piecewise linear functions?

A $f(x) = \min\{10x - 1,\ 2x + 10,\ 20\}, \quad x \in \mathbb{R}$

B $f(x) = \sum_{i=1}^{2019} 2^{-i} |e^{i/2}x - \sin(i)|, \quad x \in \mathbb{R}$

C $f(x_1, x_2) = -|x_1| + |x_2|, \quad 0 \le x_1 \le 2, 0 \le x_2 \le 2.$

D None of the above.

Solution: B or C A is not correct because the minimum of linear functions is concave, not convex. B is correct because each term of the sum can be reformulated as the maximum of two linear (i.e. convex) functions which is itself convex, and the sum of convex functions is convex. C is correct because over the given range of x_1 and x_2 values this function is equivalent to the function $-x_1+x_2$, which is linear and thus convex.

Problem 19. Choose the correct statement about the "here-and-now" decision and the "wait-and-see" decision.

A The "wait-and-see" decision can be made before part of the uncertainty is realized.

B The "here-and-now" decision may change after the uncertainty is realized.

C The "here-and-now" decision must be made before the uncertainty is realized.

Solution: C A is incorrect because all uncertainty must be realized before a "wait-and-see" decision can be made. B is not correct precisely because of what is said in C.

Problem 20. Consider the following nonlinear optimization problem:

$$\begin{aligned} \min\ & |x-2| + 3|1-x| \\ \text{s.t.}\ & x \ge 0 \end{aligned}$$

Which of the following linear reformulations is equivalent to it?

A

$$
\begin{aligned}
\min\ & z_1 + 3z_2 \\
\text{s.t.}\ & z_1 \geq x - 2 \\
& -z_1 \leq x - 2 \\
& z_2 \geq 3(1 - x) \\
& -z_2 \leq 3(1 - x) \\
& x \geq 0
\end{aligned}
$$

B

$$
\begin{aligned}
\min\ & z_1 + 3z_2 \\
\text{s.t.}\ & z_1 \geq x - 2 \\
& -z_1 \geq x - 2 \\
& z_2 \geq 1 - x \\
& -z_2 \geq 1 - x \\
& x \geq 0
\end{aligned}
$$

C

$$
\begin{aligned}
\min\ & z_1 + 3z_2 \\
\text{s.t.}\ & z_1 \geq x - 2 \\
& -z_1 \leq x - 2 \\
& z_2 \geq 1 - x \\
& -z_2 \leq 1 - x \\
& x \geq 0
\end{aligned}
$$

D

$$
\begin{aligned}
\min\ & -z_1 - 3z_2 \\
\text{s.t.}\ & z_1 \geq x - 2 \\
& -z_1 \leq x - 2 \\
& z_2 \geq 1 - x \\
& -z_2 \leq 1 - x \\
& x \geq 0
\end{aligned}
$$

Solution: C In a minimization setting, any term of the form $|x|$ in the objective can be replaced with a z where $z \geq x$ AND $z \geq -x$, which is precisely what happens in C (along with multiplying both sides of the second and fourth inequalities by -1 to mix things up a bit).

Problem 21. Consider the following nonlinear objective that is the average of the ℓ_1 metric and ℓ_∞ metric

$$\min_x \left\{ \frac{1}{2}(|x-4| + |x+5|) + \frac{1}{2}\max\{|x-4|, |x+5|\} \right\}.$$

Which is the correct reformulation as a linear program?

A

$$\begin{aligned} \min \quad & \frac{1}{2}(z_1 + z_2) + \frac{1}{2}(x-4) + \frac{1}{2}(x+5) \\ \text{s.t.} \; & -z_1 \leq x - 4 \\ & z_1 \leq x - 4 \\ & -z_2 \leq x + 5 \\ & z_2 \leq x + 5 \end{aligned}$$

B

$$\begin{aligned} \min \quad & \frac{1}{2}(z_1 + z_2) + \frac{1}{2}z_3 + \frac{1}{2}z_4 \\ \text{s.t.} \; & -z_1 \leq x - 4 \leq z_1 \\ & -z_2 \leq x + 5 \leq z_2 \\ & -z_3 \leq x - 4 \leq z_3 \\ & -z_4 \leq x + 5 \leq z_4 \end{aligned}$$

C

$$\begin{aligned} \min \quad & \frac{1}{2}z_1 + \frac{1}{2}z_2 \\ \text{s.t.} \; & -z_1 \leq x - 4 \leq z_1 \\ & -z_1 \leq x + 5 \leq z_1 \\ & -z_2 \leq x - 4 \leq z_2 \\ & -z_2 \leq x + 5 \leq z_2 \end{aligned}$$

D

$$\begin{aligned} \min \quad & \frac{1}{2}(z_1 + z_2) + \frac{1}{2}z_3 \\ \text{s.t.} \; & -z_1 \leq x - 4 \leq z_1 \\ & -z_2 \leq x + 5 \leq z_2 \\ & -z_3 \leq x - 4 \leq z_3 \\ & -z_3 \leq x + 5 \leq z_3 \end{aligned}$$

Solution: D When $-z_1 \leq x - 4 \leq z_1$ and $-z_2 \leq x + 5 \leq z_2$ are satisfied, in the minimizing problem, z_1 is equivalent to $|x-4|$ and z_2 is equivalent to $|x+5|$. For the second part of the objective function, since we need to bound both $|x-4|$ and $|x+5|$ at the same time, we use the same variable z_3.

Problem 22. Which of the following statements about the set $P = \{\boldsymbol{x} = (x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 : x_1 + 2x_2 - 3x_3 + 4x_4 - 5x_5 = 0\}$ in $\mathbb{R}^5$ is TRUE?

A P is not a polyhedron, because a polyhedron should be defined by inequalities.

B P is a polyhedron but not convex.

C P is unbounded and is the intersection of two halfspaces, $\{\boldsymbol{x} \in \mathbb{R}^5 : x_1+2x_2-3x_3+4x_4-5x_5 \leq 0\} \cap \{\boldsymbol{x} \in \mathbb{R}^5 : x_1 + 2x_2 - 3x_3 + 4x_4 - 5x_5 \geq 0\}$.

D P has an extreme ray $(3, 0, 1, 0, 0)^\top$.

Solution: C Since P can be described by finite number of inequalities, it is a polyhedron. Then, A is not true. Since all the polyhedrons are convex, B is not true. Since P contains lines, it does not have any extreme ray. Then, D is not true. Since an equality can be regard as two inequalities as described in C, C is true.

Problem 23. If a standard form linear program is not degenerate and has a finite optimal solution, then the step size determined by the min-ratio test in each iteration of the simplex method must be strictly positive, and the objective function value must also be strictly improved at each iteration.

A True

B False

Solution: A When the linear program is not degenerate, simplex method will never get stuck, which means we get a better objective value from each iteration, ie. the objective value is strictly improved.

Problem 24. Find the extreme ray(s) of the polyhedron $P = \{(x, y) \in \mathbb{R}^2 : x - y \leq 1,\ x \geq 0,\ y \geq 0\}$.

A $[1, 1]^\top$

B $[1, 0]^\top$

C The polyhedron P does not have any extreme ray.

Solution: A This is identical to the M13L3 knowledge check. Graphing the feasible region indicates that A is an extreme ray and B is not.

Problem 25. Which of the following is TRUE?

A A polyhedron always contains at least one extreme ray.

B A nonempty polytope may not have a extreme point.

C The nonnegative orthant $\{(x_1, x_2, x_3) : x_i \geq 0, \forall i = 1, 2, 3\}$ has three extreme rays.

D A nonempty polyhedron in standard form may not have an extreme point.

Solution: C For A, $\mathbb{R}^2$ has no extreme rays. For B, since every polytope is bounded, it must have a vertex, ie. an extreme point. For D, we can always find a basic solution. Then, it must have an extreme point. C is true.

Problem 26. Given a polyhedron P in standard form where A is a 4×6 matrix with full row rank, which of the following is not possible to be the number of basic feasible solutions of P?

A 20

B 15

C 1

D 0

Solution: A Since A is a 4×6 matrix with full row rank, $x \in \mathbb{R}^6$ has 4 basic variables. We have at most 15 ways to choose 4 from 6. Therefore, at most 15 basic feasible solutions of P.

Problem 27. Suppose a standard form LP has a nondegenerate basic feasible solution $\boldsymbol{x} = (x_1 = 1, x_2 = 2, x_3 = 0, x_4 = 2, x_5 = 0)$. Which of the following points cannot be a basic feasible solution adjacent to $\boldsymbol{x}$?

A $(2, 1, 3, 0, 0)$.

B $(0, 1, 0, 1, 3)$.

C $(0, 0, 0, 1, 1)$.

D $(1, 0, 2, 0, 1)$.

Solution: D By the definition of two adjacent basic feasible solutions, only one nonbasic variable can be different between them. Since D has two different nonbasic variables, D cannot be a basic feasible solution adjacent to x.

Problem 28. Suppose the direction to move in a simplex iteration is $\boldsymbol{d}^\top = \begin{pmatrix} -3 & 0 & -4 & 1 & 0 \end{pmatrix}$ and the current basic variables have values $(1, 2)$. Which variables are basic variables and what is the next iteration's basic feasible solution?

A Basic variables are (x_2, x_5) and the next iteration's basic feasible solution is $\begin{pmatrix} 0 & 0 & -2 & 1 & 0 \end{pmatrix}^\top$.

B Basic variables are (x_1, x_3) and the next iteration's basic feasible solution is $\begin{pmatrix}1 & 0 & 0 & \frac{1}{2} & 0\end{pmatrix}^\top$.

C Basic variables are (x_1, x_3) and the next iteration's basic feasible solution is $\begin{pmatrix}0 & 0 & \frac{2}{3} & \frac{1}{3} & 0\end{pmatrix}^\top$.

Solution: C A is not correct since it includes a negative-valued variable and we only use simplex on a problem in standard form. Note that x_4 must be the entering variable since it's the only variable with its corresponding entry in d equal to 1, and that x_1 and x_3 are basic since their corresponding entries of d are nonzero. To differentiate between B and C, we need to use the ratio test to determine how much we can increase the entering variable x_4. The ratio for x_1 is $\frac{1}{3}$, which is smaller than the ratio of $\frac{1}{2}$ for x_3. Thus x_1 is first to leave the basis, meaning C is correct.

Problem 29. If the the reduced costs of nonbasic variables (x_3, x_4, x_5) are $(3, -1, 0)$ and the ratios are $(\frac{x_1}{-d_1}, \frac{x_2}{-d_2}) = (1, 0)$. Is this basic feasible solution degenerate and why?

A Yes, because currently the reduced cost of x_5 is 0.

B No, because no basic variable can be zero.

C Yes, because the basic variable x_2 must be 0.

Solution: C For the min-ratio test, we only consider entries in d that are strictly less than 0, meaning that in order for the ratio $\frac{x_2}{-d_2}$ to evaluate to 0, it must be the case that $x_2 = 0$. Since x_2 is basic, we have a degenerate BFS.

Problem 30. Given an LP , write down an equivalent standard form LP.
$\max\{2x_1 - 3x_2 : x_1 + x_2 \leq 2,\ x_1 + 2x_2 \geq 3\}$

A $\max\{2x_1 - 3x_2 \ :\ x_1 + x_2 + s_1 = 2,\ x_1 + 2x_2 - s_2 = 3, s_1 \geq 0, s_2 \geq 0\}$

B $\min\{-2x_1^+ + 2x_1^- + 3x_2 \ :\ x_1^+ - x_1^- + x_2 + s_1 = 2,\ x_1^+ - x_1^- + 2x_2 - s_2 = 3, s_1 \geq 0, s_2 \geq 0, x_1^+ \geq 0, x_1^- \geq 0\}$

C $\min\{-2x_1^+ + 2x_1^- + 3x_2^+ - 3x_2^- \ :\ x_1^+ - x_1^- + x_2^+ - x_2^- + s_1 = 2,\ x_1^+ - x_1^- + 2x_2^+ - 2x_2^- - s_2 = 3, s_1 \geq 0, s_2 \geq 0, x_1^+ \geq 0, x_1^- \geq 0, x_2^+ \geq 0, x_2^- \geq 0\}$

D $\max\{2x_1^+ - 2x_1^- - 3x_2^+ + 3x_2^- \ :\ x_1^+ - x_1^- + x_2^+ - x_2^- + s_1 = 2,\ x_1^+ - x_1^- + 2x_2^+ - 2x_2^- - s_2 = 3, s_1 \geq 0, s_2 \geq 0, x_1^+ \geq 0, x_1^- \geq 0, x_2^+ \geq 0, x_2^- \geq 0\}$

Solution: C Since both variables are unrestricted, we need to split them into their positive and negative parts so that all variables can be non-negative. We also need to negate the objective function, add a non-negative slack variable to the first constraint, and subtract a non-negative slack variable from the second constraint.